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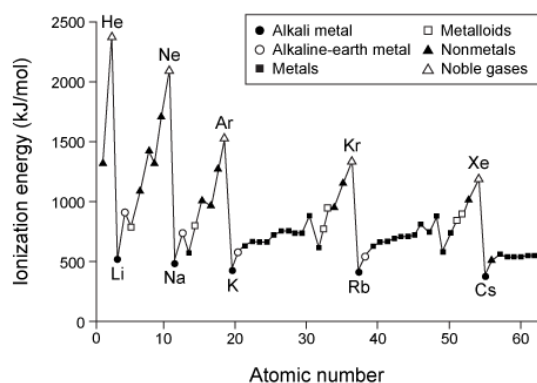


Figure 1 First ionization energy of elements

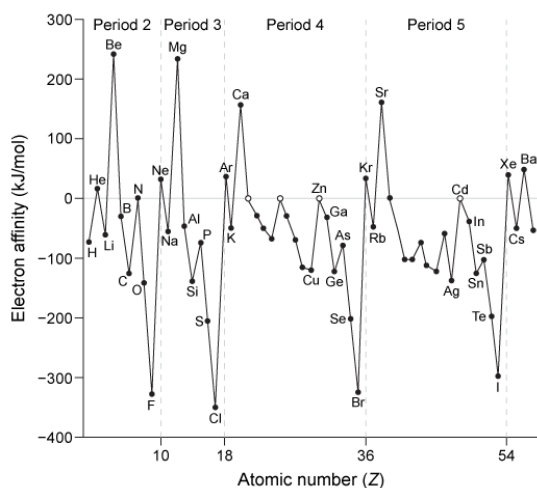


Figure 2 Electron affinity of several elements (larger negative values indicate greater electron affinity)

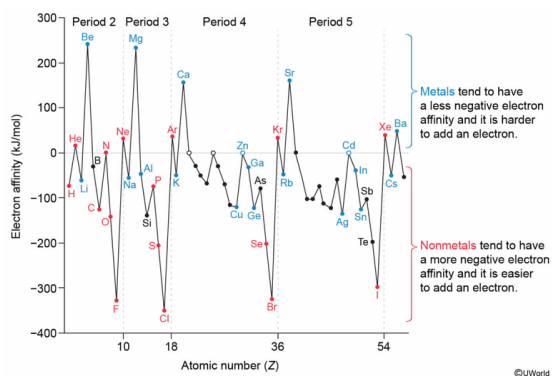
According to the information in the passage, in general, adding electrons to nonmetals is:

- ☐ A. not possible for most nonmetals. [1%]
- ☐ B. energetically similar to adding electrons to metals. [3%]
- ✓ ☒ C. easier than adding electrons to metals. [69%]
- ☐ D. more difficult than adding electrons to metals. [25%]

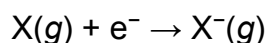
Correct

69% answered correctly

Explanation:



Electron affinity is defined as the **change in energy** resulting from **adding an electron** to a neutral atom of an element X in the gas state to form an anion with a -1 charge:



When a stable anion is formed, energy is released, which results in a negative value for the change in energy. Therefore, elements with a **more negative** electron affinity **more readily accept** the addition of an **electron**. **Electron affinity trends** show that apart from some exceptions caused by atomic size and orbital-filling effects (eg, nitrogen and the noble gases), values tend to become more negative (higher affinity) moving left to right across a period and become more positive (lower affinity) moving down a group on the periodic table.

The data in Figure 2 indicates that nonmetals tend to have the most negative values for electron affinity. Therefore, **adding an electron** to a **nonmetal** is generally **easier** than adding an electron to a metal.

(Choice A) Adding an electron to the valence shell is possible for most atoms, although some metals may require the addition of energy to do so.

(Choice B) In general, the energy change from adding an electron to a nonmetal (such as chlorine) is very different from that of adding an electron to a metal (such as barium). Chlorine releases energy (-350 kJ/mol) whereas barium *requires* energy ($+50$ kJ/mol), according to

Figure 2.

(Choice D) The more negative the electron affinity, the easier it is to add an electron to the valence shell of an atom. In general, the electron affinity of nonmetals is more negative than that of metals.

Educational objective:

Electron affinity is the change in energy associated with adding an electron to a neutral atom in the gas state. The more negative the electron affinity, the more readily an atom accepts the addition of an electron. Most nonmetals tend to accept an additional electron more readily than most metals.

Time spent:0 sec**Subject:**
General Chemistry**QID:**401407**System:**
Atomic Theory and Chemical Composition

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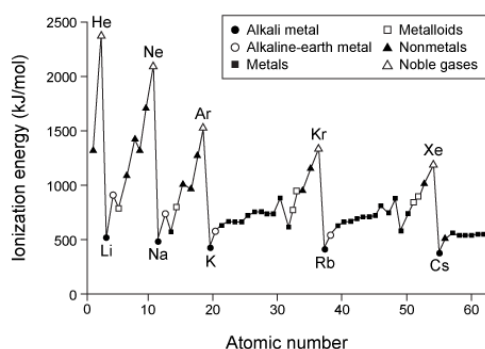


Figure 1 First ionization energy of elements

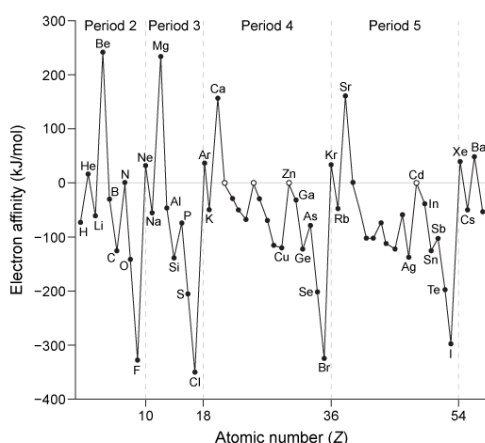


Figure 2 Electron affinity of several elements (larger negative values indicate greater electron affinity)

The difference between the first and second ionization energies for magnesium is approximately 700 kJ/mol. The difference between the second and third ionization energies will likely be:

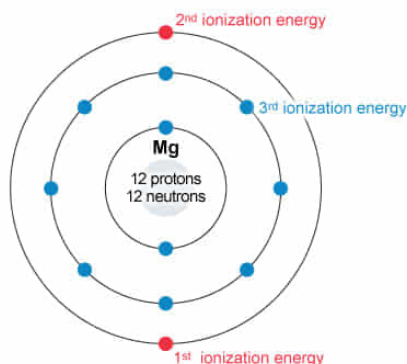
- ☐ A. higher because the first and second ionization energies remove electrons from the *p* orbitals whereas the third removes an electron from an *s* orbital. [7%]
- ☒ B. higher because the first and second ionization energies remove valence electrons whereas the third removes a core electron. [74%]

- ☐ C. lower because the first and second ionization energies remove spin-paired electrons whereas the third removes an electron that is not spin-paired. [7%]
- ☐ D. lower because the first and second ionization energies remove electrons from the $n = 2$ shell whereas the third removes an electron from the $n = 3$ shell. [9%]

Correct

74% answered correctly

Explanation:



In Mg, 1st and 2nd ionization energies remove valence electrons, and the 3rd ionization energy removes a core electron

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Ionization energy is the energy needed to **remove an electron** from a ground state gaseous atom or ion. The first, second, and third ionization energies are the energies required to remove the first, second, and third electron, respectively. Because the **valence shell** is farther away from the positively charged nucleus, electrons in this shell are easier to remove than the inner core electrons. The electrons are removed in order beginning with the **electron of highest energy**.

Magnesium has **two electrons** in its valence shell ($n = 3$). Magnesium's third ionization energy is significantly larger because it requires the removal of an electron from the **inner core**. These electrons are **more tightly held** than the valence electrons, and therefore require more energy to remove.

(Choice A) Although orbital shape does have some effect on ionization energy (eg, s electrons are more tightly held than p electrons), energy differences based on the distance from the nucleus have a much larger effect. In the case of magnesium, the first and second electrons are removed from an s orbital, not a p orbital.

(Choice C) "**Spin-paired**" electrons occur when two electrons occupy the same atomic orbital and have opposite spins. Magnesium has two electrons in its valence shell that are spin-paired in the 3s orbital; the third electron is also spin-paired to another electron in one of the 2p orbitals.

(Choice D) Electrons are **removed** beginning with the shell that is farthest from the nucleus. In the case of magnesium, electrons are removed from the $n = 3$ shell first and then from the $n = 2$ shell.

Educational objective:

Ionization energy is the energy required to remove an electron from a ground-state gaseous atom or ion. The first, second, and third ionization energies describe the energy required to remove one, two, or three electrons, beginning with the electron of highest energy. Electrons in the inner core require significantly more energy to remove than valence electrons.

Time spent: 0 sec

Subject:
General Chemistry

QID: 401408

System:
Atomic Theory and Chemical Composition

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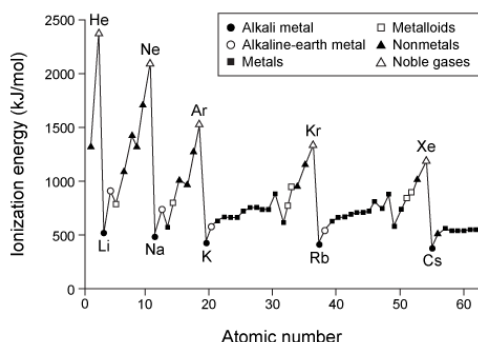


Figure 1 First ionization energy of elements

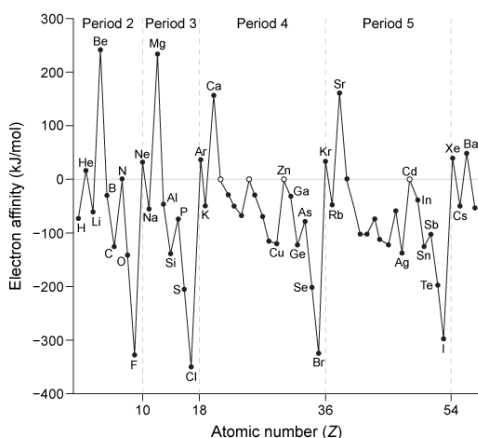


Figure 2 Electron affinity of several elements (larger negative values indicate greater electron affinity)

Why are noble gases the least reactive group on the periodic table?

- ☐ A. Noble gases are diamagnetic. [1%]
- ✓ ☒ B. Noble gases have a full valence shell. [97%]
- ☐ C. Noble gases tend to have the same number of protons and neutrons.
- ☐ D. Noble gases tend to form diatomic molecules. [1%]

Correct

96% answered correctly

Explanation:

Valence shell drives chemistry

Atoms tend to gain, lose, or share electrons to achieve a full valence shell

1e ⁻ lose	2e ⁻ lose	<valence electrons> <most likely change>	3e ⁻ lose	4e ⁻ either	5e ⁻ gain	6e ⁻ gain	7e ⁻ gain	8e ⁻ none
1A H•	2A Li• Be•	Transition metals may have various charges	3A •B•	4A •C•	5A •N•	6A •O•	7A •F•	8A He•
Na•	Mg•		•Al•	•Si•	•P•	•S•	•Cl•	•Ar•
K•	Ca•							

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Noble gases have **full s and p orbitals** in their **valence shells** except for helium, which has only a full s orbital in its valence shell. For each energy level n , the s orbital can house two electrons and the p orbitals six electrons, for a total of eight electrons. Because the valence orbitals are filled, it is not possible to accept another electron at that energy level, and it is energetically unfavorable to accept more electrons at higher levels.

Given that reactivity is driven by the tendency of atoms to gain, lose, or share electrons to achieve a **more stable electronic configuration**, noble gases tend to be **unreactive** (except in extreme cases) as they already possess an energetically favorable configuration.

(Choice A) Although it is true that noble gases are **diamagnetic** (no unpaired electrons), having paired electrons does not mean that an atom is unreactive. For example, magnesium has no unpaired electrons but it reacts readily with several Group 6A and Group 7A elements.

(Choice C) The number of protons and neutrons does not determine an atom's reactivity. Atoms are reactive based on their electron configuration.

(Choice D) Diatomic elements naturally occur in a covalently bonded pair, which allows each atom in the pair to achieve a full valence shell. For example, hydrogen is typically found as H_2 and oxygen is typically found as O_2 . However, noble gases are *not* diatomic because they already have a full valence shell of electrons.

Educational objective:

Reactivity is driven by the tendency of atoms to gain, lose, or share electrons to achieve a more stable electronic configuration (ie, a full valence shell). Noble gases are generally unreactive because they already have a full valence shell of electrons.

Time spent: 0 sec

Subject:
General Chemistry

QID: 401409

System:
Atomic Theory and Chemical Composition

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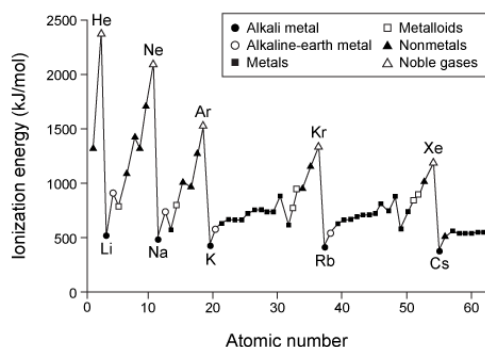


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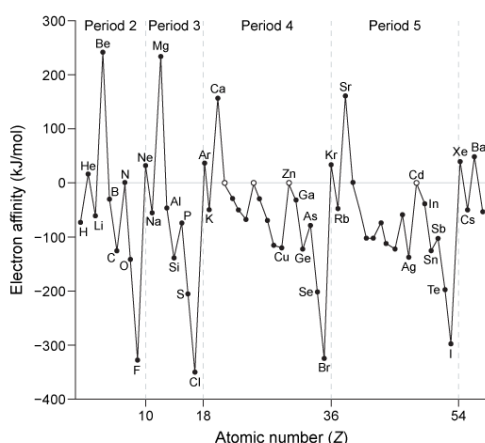


Figure 2 Electron affinity of several elements (larger negative values indicate greater electron affinity)

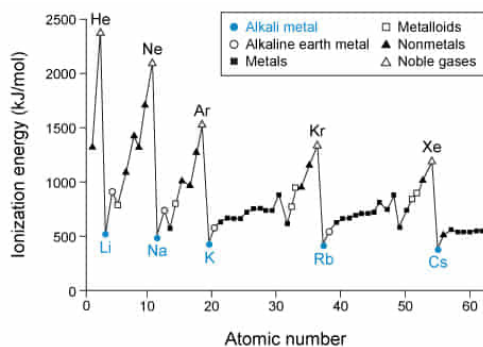
According to trends in ionization energy, which metal would be the most reactive with Group 6A and 7A nonmetals?

- ☐ A. Li [30%]
- ☐ B. Na [3%]
- ☐ C. K [4%]
- ☒ D. Cs [62%]

Correct

61% answered correctly

Explanation:



Alkali metals have low ionization energies.
Cs has the **lowest ionization energy** and is therefore the **most reactive alkali metal**.

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Alkali metals, or **Group 1** metals, are known to be **very reactive** because of their **low first ionization energy**, large **atomic radius**, and **small electronegativity**. The alkali metals have **one electron** in their valence shell, which they readily donate to form strong ionic bonds with nonmetals such as the **halogens** (Group 7A, or Group 17). As the atoms in the alkali metal group increase in size, the valence electrons occupy **higher energy levels**, and less energy is required to remove an electron from the valence shell. Therefore, larger alkali metals are more reactive.

Cesium (Cs) has the lowest first ionization energy of all the elements. Consequently, less energy is required to remove an electron from its valence shell compared to other metals. As a result, **Cs** is the **most reactive metal** in reactions with nonmetals.

(Choices A, B, and C) Lithium (Li), sodium (Na), and potassium (K) are all alkali metals, but each has a larger ionization energy than Cs. For reactions with nonmetals, more energy is required to remove an electron from Li, Na, or K than from Cs, making these alkali metals less reactive than Cs.

Educational objective:

The alkali metals are very reactive because of their low ionization energy, large atomic radius, and small electronegativity. Cesium is considered the most reactive metal.

Time spent: 0 sec

Subject:

General Chemistry

QID: 401410

System:

Atomic Theory and Chemical Composition

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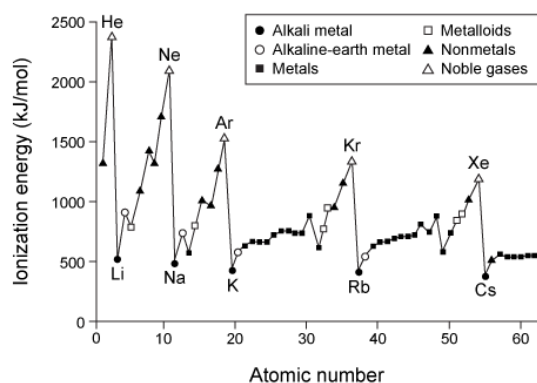


Figure 1 First ionization energy of elements

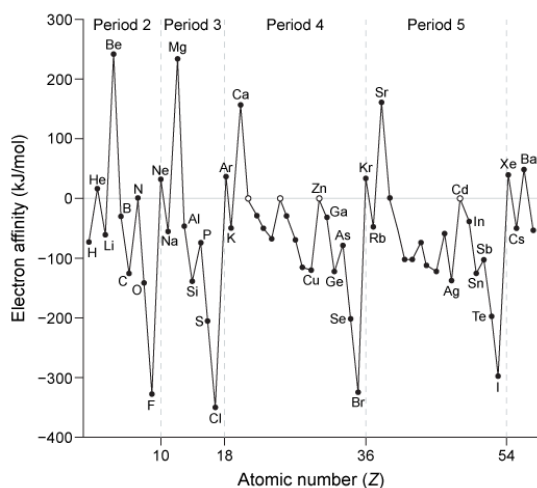
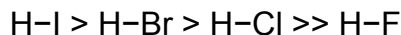


Figure 2 Electron affinity of several elements (larger negative values indicate greater electron affinity)

The acid strength of hydrogen halides is ranked from strongest to weakest as follows:



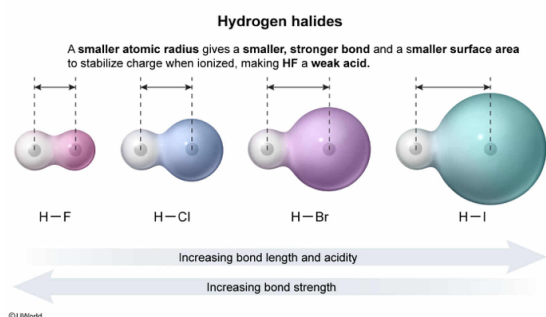
Which of the following best explains why H-F is not a strong acid?

- ☐ A. Fluorine has the largest atomic radius, making the H-F bond weak. [5%]
- ☐ B. Fluorine has the largest atomic radius, making the H-F bond strong. [1%]
- ☐ C. Fluorine has the smallest atomic radius, making the H-F bond weak. [14%]
- ✓ ☒ D. Fluorine has the smallest atomic radius, making the H-F bond strong. [78%]

Correct

78% answered correctly

Explanation:



The **size** of the **halogen atom** impacts the bond length and strength of hydrogen halide acids (H-X). Halogens are located in Group 7A (Group 17) on the right side of the [periodic table](#) and increase in size moving down the group. Therefore, halogens with a smaller atomic number have a **smaller atomic radius** and form shorter, **stronger H-X bonds** that are more difficult to ionize. Smaller halogens are more [electronegative](#) and more readily attract electrons in the H-X bond toward itself, creating a polar covalent bond.

The stability of the [conjugate base](#) of the hydrogen halides (X⁻) also affects acidity. The conjugate base formed from a smaller halogen is less stable because it has a smaller atomic radius (less surface area) and cannot spread out (stabilize) the negative charge as well as halogens with larger atomic radii (greater surface area).

Fluorine is the most electronegative halogen and has the smallest atomic radius of the halogens. Therefore, H-F forms the shortest, strongest bond of all the hydrogen halides. A short, strong H-X bond that is difficult to ionize in aqueous solution indicates a [weak acid](#).

(Choices A and B) Fluorine has the *smallest* atomic radius of the halogens rather than the largest.

(Choice C) Although fluorine does have the smallest atomic radius of the halogens, the H-F bond is *strong* rather than weak. This strong bond is due to fluorine having a small atomic radius.

Educational objective:

The acid halides are all strong acids except for H-F. Fluorine is the most electronegative and

has the smallest atomic radius of the halides, resulting in strong bond formation.

Time spent:0 sec

Subject:
General Chemistry

QID:401411

System:
Atomic Theory and Chemical Composition

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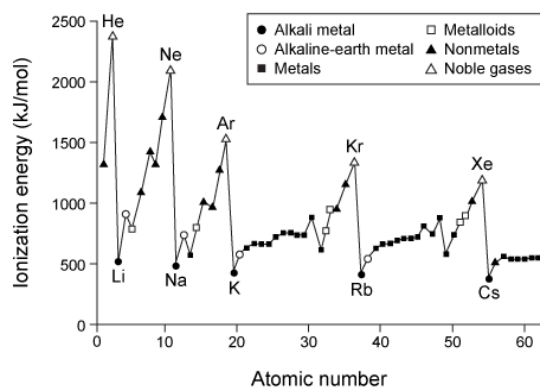


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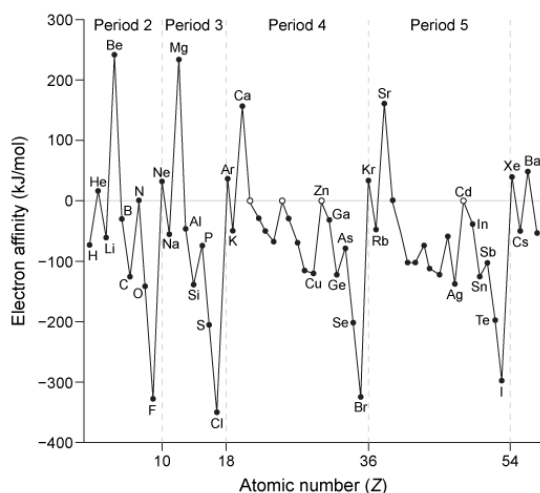


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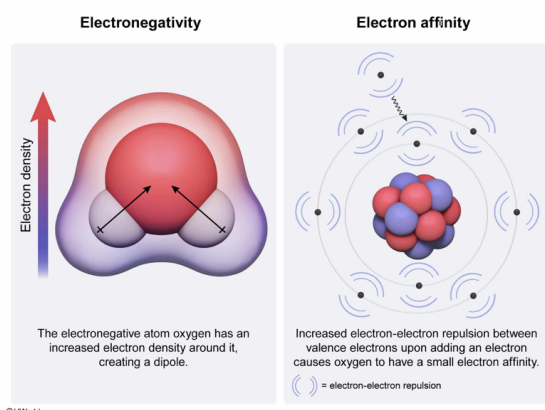
Oxygen is the most electronegative element of Group 6A. However, it also has the least negative electron affinity in the group. What factor can be used to explain this discrepancy?

- ☐ A. Oxygen is less stable (more reactive) than the other Group 6A elements. [24%]
- ✓ ☒ B. Oxygen's valence electrons experience more electron-electron repulsion than other Group 6A elements. [39%]
- ☐ C. Oxygen's valence electrons are more shielded from the positively charged nucleus than other Group 6A elements. [16%]
- ☐ D. Oxygen does not have any partially-filled *d* orbitals in its valence shell for additional electrons. [19%]

Correct

39% answered correctly

Explanation:



Electronegativity is a qualitative measure of how strongly electrons are attracted to an atom *within* a bond. Group 6A (Group 16) elements follow the trend that **electronegativity** decreases going down the group on the periodic table. When bonded to another atom, oxygen strongly attracts electrons toward itself, creating unequal electron sharing that produces a **dipole moment**.

In contrast, **electron affinity** (EA) is the quantitative measure of the **energy change** that occurs when an electron is added to an atom in the gas state. The more negative the change in energy, the more energy released upon **addition of an electron**. Although EA generally increases (becomes more negative) from left to right across a period, EA generally *decreases* (becomes more positive) going down a group because the electron is added at an increased distance from the nucleus.

The exception to this trend is when an atom is very small (ie, first or second period). Small atoms such as oxygen and fluorine have several electrons crowded around a small nucleus, resulting in greater electron-electron repulsion. Adding another electron would increase repulsion forces even further. Therefore, it is more difficult to add an electron to oxygen than to a larger atom with lower repulsion forces.

(Choice A) Although oxygen is less stable (more reactive) than other Group 6A elements,

this does not explain why its electron affinity breaks the pattern.

(Choice C) Oxygen's valence electrons are *less* shielded from the nucleus than other Group 6A elements because its valence electrons are closer to the nucleus.

(Choice D) Although it is true that oxygen does not have any partially-filled *d* orbitals in its valence shell, this does not explain why oxygen has a less negative electron affinity than expected. Sulfur, which has a *more negative* electron affinity than oxygen, also does not have partially-filled *d* orbitals in the ground state of its valence shell.

Educational objective:

Electron affinity is the measure of energy change when an electron is added to an atom in the gaseous state. Electron-electron repulsions result in a decreased (less negative) electron affinity.

Time spent:0 sec

Subject:
General Chemistry

QID:401412

System:
Atomic Theory and Chemical Composition